Ex.no:8	MINI PROJECT – HOTEL BOOKINGS DATA ANALYTICS

Domain and Problem Statement:

Domain: Tourism Analytics & Destination Management

Problem Statement: Tourism organizations need to understand destination preferences, travel patterns, estimated budgets, visitor volumes, ratings, and seasonal demand. This project transforms tourism records into structured business intelligence using Python, MySQL, Power BI, and Tableau.

Dataset Description:

Dataset Name: Tourism Dataset .

Granularity: Each row represents a tourism destination record .

Total Records: 200 .

Key Entities & Attributes:

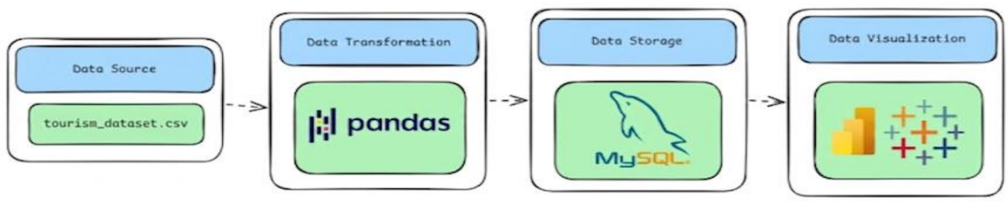
- **Identification**: Tourism_ID, Destination.
- **Geography**: Country, Continent.
- **Tourism Characteristics:** Tourism_Type, Best_Season.
- **Trip Characteristics**: Average_Duration_Days, Estimated_Budget_USD, Rating.
- **Demand**: Annual_Visitors.
- **Travel Services**: Accommodation_Type, Transportation.

Data Source:

Primary Source: Kaggle Open Datasets .

Storage Infrastructure: The CSV can be processed locally with Python and, if required, loaded into a local MySQL database for Power BI and Tableau connectivity.

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd

from sqlalchemy import create_engine

df = pd.read_csv("tourism_dataset_200.csv")

tourism = df.copy()

tourism["Rating"] = tourism["Rating"].fillna(tourism["Rating"].median())

tourism["Estimated_Budget_USD"] = tourism["Estimated_Budget_USD"].fillna(

tourism["Estimated_Budget_USD"].median()

)

tourism["Tourism_Type"] = tourism["Tourism_Type"].fillna(

tourism["Tourism_Type"].mode()[0]

)

tourism.drop_duplicates(inplace=True)

tourism["Budget_Per_Day_USD"] = (

tourism["Estimated_Budget_USD"] /

tourism["Average_Duration_Days"]

).round(2)

tourism["Visitor_Category"] = pd.cut(

tourism["Annual_Visitors"],

bins=[0, 100000, 300000, float("inf")],

labels=["Low", "Medium", "High"]

)

tourism.columns = (

tourism.columns.str.lower().str.strip().str.replace(" ", "_")

)

print("Original Shape:", df.shape)

print("Cleaned Shape:", tourism.shape)

username = "root"

password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
port = 3306
db_engine = create_engine(
f"mysql+pymysql://{username}:{password}@{host}:{port}/tourism_analytics"
)
tourism.to_sql(
"tourism_data",
con=db_engine,
if_exists="replace",
index=False,
chunksize=500
)
print("Data loaded successfully into MySQL.")
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80
mysql -u root -p
```

MySQL:

```
CREATE DATABASE IF NOT EXISTS tourism_analytics;
```

Python/Jupyter:

```
username = "root"
password = "YOUR_MYSQL_PASSWORD"
host = "localhost"
port = 3306

# Connect to MySQL database
db_engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/tourism_analytics"
)
```

Load cleaned data into MySQL

```
tourism.to_sql(
 "tourism_destinations",
 con=db_engine,
 if_exists="replace",
 index=False
)
print("Data loaded successfully into MySQL.")
```

Verify:

SELECT COUNT(*) AS total_rows FROM tourism_data;

Procedure:

Step 1: Data Preparation & Preprocessing (Python / Pandas)

- Load tourism_dataset_200.csv into a Jupyter Notebook or Python script using pandas.read_csv().
- Inspect data types, missing values, duplicate records, and unusual numeric values.
- Remove duplicate destination records where appropriate and fill missing categorical and numeric values.
- Create Budget_Per_Day_USD and Visitors_Millions as derived analytical fields.
- Export the cleaned DataFrame for database loading and BI visualization.

```
Original rows: 200
Cleaned rows: 200
Columns: 14

Missing values:
Series([], dtype: int64)
```

	Tourism_ID	Destination	Country	Continent	Tourism_Type	Best_Season	Average_Duration_Days	Estimated_Budget_USD	Rating	Annual_Visitors	Accommodation_Type	Transport
0	T001	Los Angeles	USA	North America	City	Winter	3	504	4.6	133393	Hotel	
1	T002	Barcelona	Spain	Europe	City	Winter	11	3756	3.5	54123	Apartment	
2	T003	Paris	France	Europe	Wildlife	Winter	10	1928	4.6	372696	Homestay	
3	T004	Singapore	Singapore	Asia	Wildlife	Spring	14	353	4.6	88707	Hostel	
4	T005	Tokyo	Japan	Asia	Cultural	Spring	14	3057	3.7	204191	Hostel	

```python
with engine.connect() as connection:
    result = connection.execute(
        text("""
            SELECT *
            FROM tourism_destinations
            LIMIT 10
        """)
    )

    for row in result:
        print(row)
```

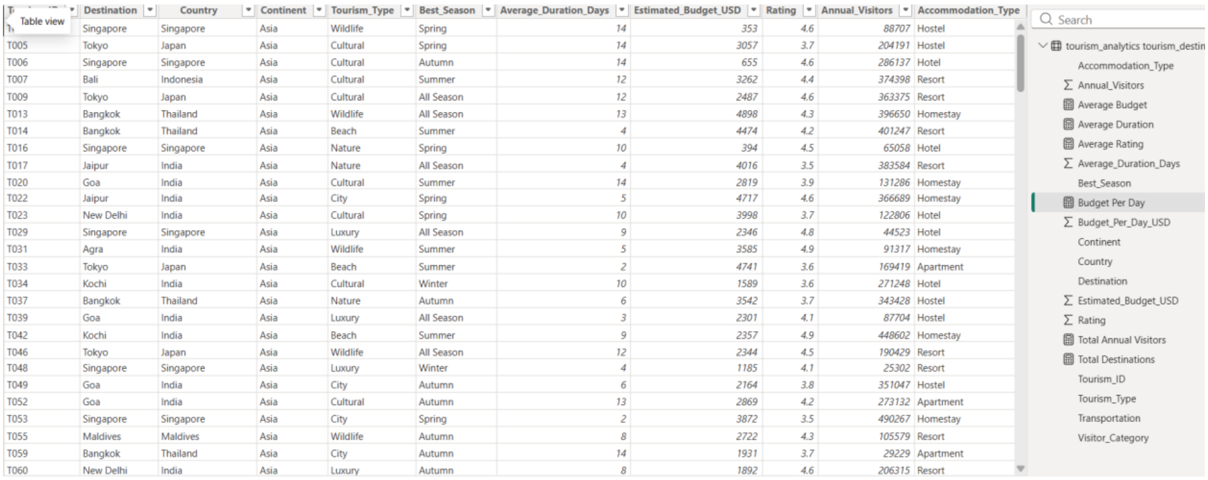
Python

```
('T001', 'Los Angeles', 'USA', 'North America', 'City', 'Winter', 3, 504, 4.6, 133393, 'Hote
('T002', 'Barcelona', 'Spain', 'Europe', 'City', 'Winter', 11, 3756, 3.5, 54123, 'Apartment'
('T003', 'Paris', 'France', 'Europe', 'Wildlife', 'Winter', 10, 1928, 4.6, 372696, 'Homestay
('T004', 'Singapore', 'Singapore', 'Asia', 'Wildlife', 'Spring', 14, 353, 4.6, 88707, 'Hoste
('T005', 'Tokyo', 'Japan', 'Asia', 'Cultural', 'Spring', 14, 3057, 3.7, 204191, 'Hostel', 'C
('T006', 'Singapore', 'Singapore', 'Asia', 'Cultural', 'Autumn', 14, 655, 4.6, 286137, 'Hote
('T007', 'Bali', 'Indonesia', 'Asia', 'Cultural', 'Summer', 12, 3262, 4.4, 374398, 'Resort',
('T008', 'Barcelona', 'Spain', 'Europe', 'Heritage', 'Winter', 5, 1127, 4.1, 242717, 'Hostel
('T009', 'Tokyo', 'Japan', 'Asia', 'Cultural', 'All Season', 12, 2487, 4.6, 363375, 'Resort'
```

Step 2: SQL Database Connection & Data Ingestion

1. Start MySQL and create a database named tourism_analytics.
2. Create/load the tourism_destinations table from the cleaned Pandas DataFrame.
3. Verify the number of records using SELECT COUNT(*) FROM tourism_destinations.

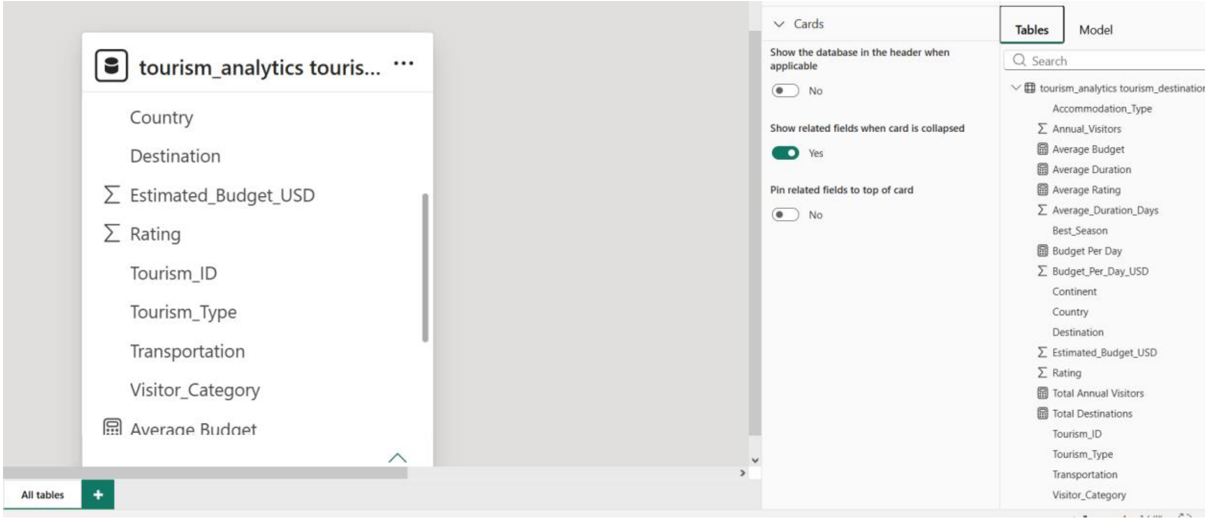
```
SELECT COUNT(*) AS total_records
FROM tourism_destinations;


SELECT Tourism_Type, COUNT(*) AS destination_count
FROM tourism_destinations
GROUP BY Tourism_Type
ORDER BY destination_count DESC;
```



[illegible]	Destination	Country	Continent	Tourism_Type	Best_Season	Average_Duration_Days	Estimated_Budget_USD	Rating	Annual_Visitors	Accommodation_Type
[illegible]	Singapore	Singapore	Asia	Wildlife	Spring	14	353	4.6	88707	Hostel
T005	Tokyo	Japan	Asia	Cultural	Spring	14	3057	3.7	204191	Hostel
T006	Singapore	Singapore	Asia	Cultural	Autumn	14	655	4.6	286137	Hotel
T007	Bali	Indonesia	Asia	Cultural	Summer	12	3262	4.4	374398	Resort
T009	Tokyo	Japan	Asia	Cultural	All Season	12	2487	4.6	363375	Resort
T013	Bangkok	Thailand	Asia	Wildlife	All Season	13	4898	4.3	396650	Homestay
T014	Bangkok	Thailand	Asia	Beach	Summer	4	4474	4.2	401247	Resort
T016	Singapore	Singapore	Asia	Nature	Spring	10	394	4.5	65058	Hotel
T017	Jaipur	India	Asia	Nature	All Season	4	4016	3.5	383584	Resort
T020	Goa	India	Asia	Cultural	Summer	14	2819	3.9	131286	Homestay
T022	Jaipur	India	Asia	City	Spring	5	4717	4.6	366689	Homestay
T023	New Delhi	India	Asia	Cultural	Spring	10	3998	3.7	122806	Hotel
T029	Singapore	Singapore	Asia	Luxury	All Season	9	2346	4.8	44523	Hotel
T031	Agra	India	Asia	Wildlife	Summer	5	3585	4.9	91317	Homestay
T033	Tokyo	Japan	Asia	Beach	Summer	2	4741	3.6	169419	Apartment
T034	Kochi	India	Asia	Cultural	Winter	10	1589	3.6	271248	Hotel
T037	Bangkok	Thailand	Asia	Nature	Autumn	6	3542	3.7	343428	Hostel
T039	Goa	India	Asia	Luxury	All Season	3	2301	4.1	87704	Hostel
T042	Kochi	India	Asia	Beach	Summer	9	2357	4.9	448602	Homestay
T046	Tokyo	Japan	Asia	Wildlife	All Season	12	2344	4.5	190429	Resort
T048	Singapore	Singapore	Asia	Luxury	Winter	4	1185	4.1	25302	Resort
T049	Goa	India	Asia	City	Autumn	6	2164	3.8	351047	Hostel
T052	Goa	India	Asia	Cultural	Autumn	13	2869	4.2	273132	Apartment
T053	Singapore	Singapore	Asia	City	Spring	2	3872	3.5	490267	Homestay
T055	Maldives	Maldives	Asia	Wildlife	Autumn	8	2722	4.3	105579	Resort
T059	Bangkok	Thailand	Asia	City	Autumn	14	1931	3.7	29229	Apartment
T060	New Delhi	India	Asia	Luxury	Autumn	8	1892	4.6	206315	Resort

Step 3: Star Schema Data Modeling (Power BI)

- Connect Power BI Desktop to the tourism_analytics MySQL database or directly to the cleaned CSV.
- Use the tourism_destinations table as the main fact table.
- Create Dim_Destination containing Tourism_ID, Destination, Country, and Continent.
- Create Dim_Tourism_Type containing unique Tourism_Type values.
- Create Dim_Season containing unique Best_Season values.
- Create Dim_Travel_Service containing Accommodation_Type and Transportation combinations.
- Relate the dimension tables to the main tourism table using the appropriate keys or category fields.
- Apply the model and organize the fields for dashboard development.



Step 4: Core Calculations & Measures (DAX)

Power BI Measures (DAX)

Create a dedicated _Measures table and add the following DAX calculations:

- **Total Destinations :**

  Total Destinations = DISTINCTCOUNT (tourism_destinations[Tourism_ID])

- **Total Annual Visitors :**

  Total Annual Visitors = SUM (tourism_destinations[Annual_Visitors])

- **Average Budget:**

  Average Budget = AVERAGE(tourism_destinations[Estimated_Budget_USD])

- **Average Rating:**

  Average Rating = AVERAGE (tourism_destinations[Rating])

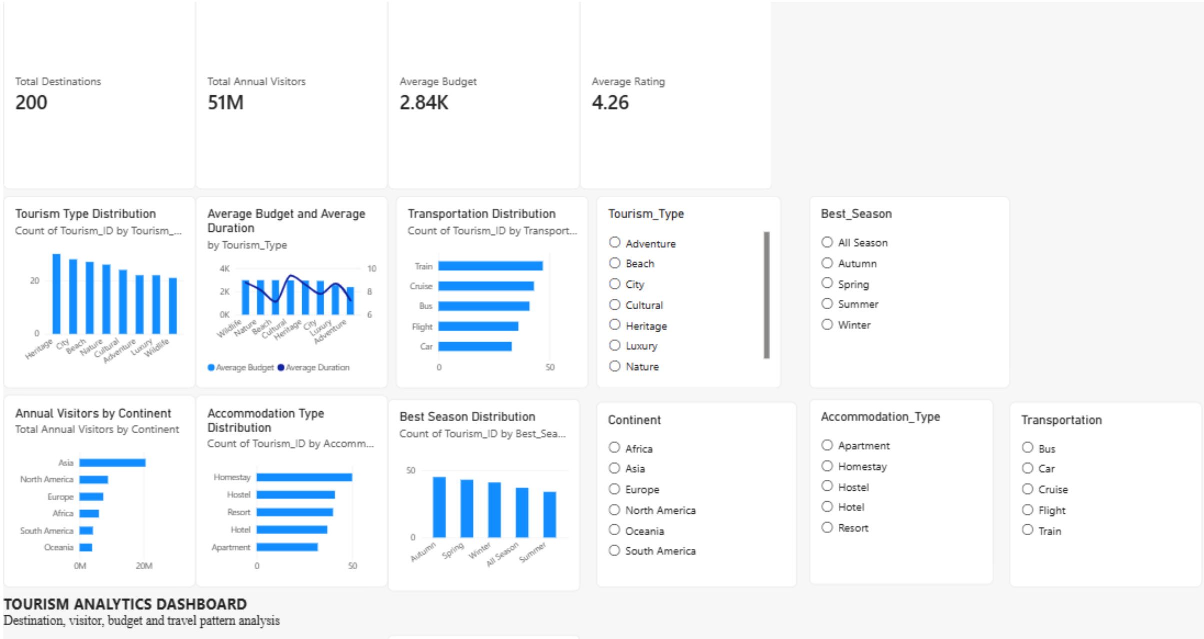
- **Average Duration:**

  Average Duration = AVERAGE(tourism_destinations[Average_Duration_Days])

- **Budget Per Day :**

```
Budget Per Day = DIVIDE (
        SUM(tourism_destinations[Estimated_Budget_USD]),
        SUM(tourism_destinations[Average_Duration_Days])
    )
```

Step 5: Power BI Final Executive Dashboard & Drill-Through



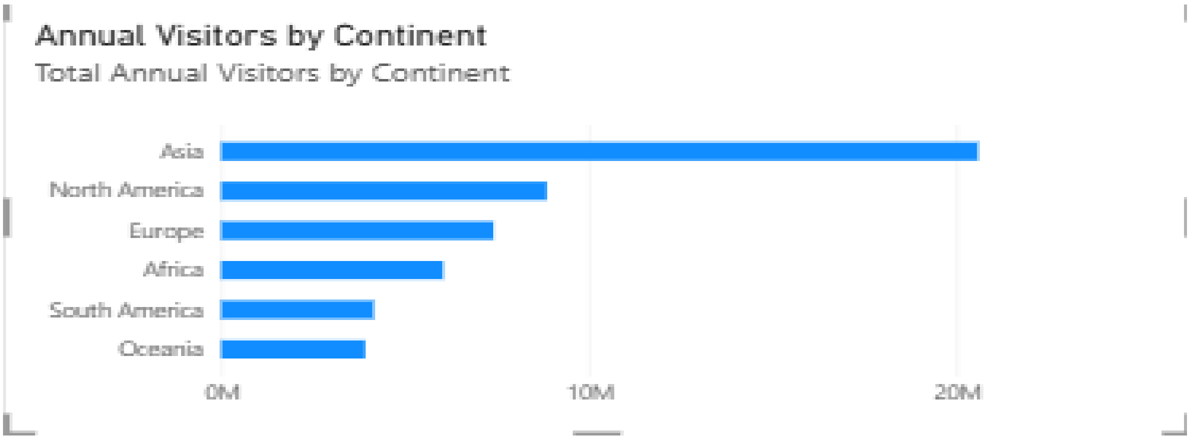
1. **Executive Canvas:**

   Place **Card Visuals** across the top: Total Destinations, Total Annual Visitors, Average Budget, Average Rating

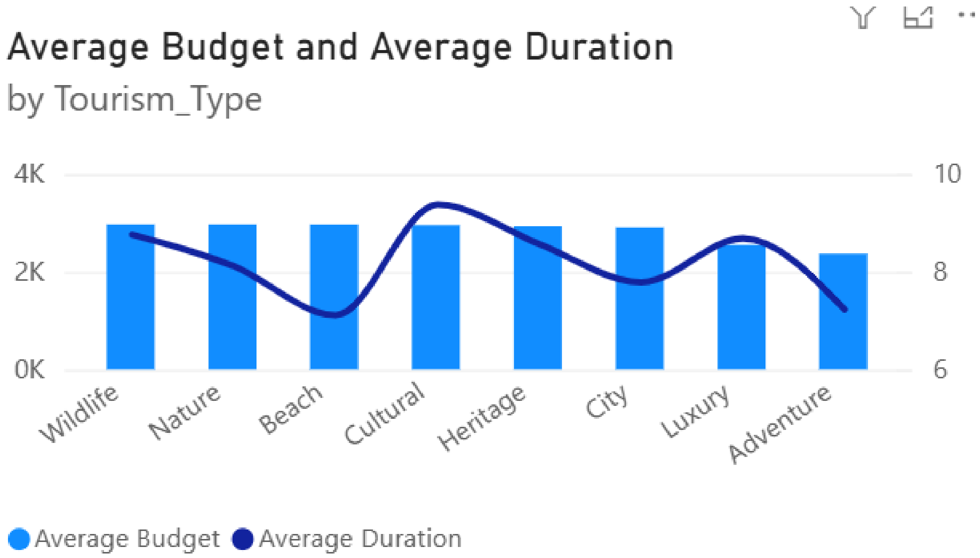
   - 

   - Add a Clustered Column Chart for Destination Count by Tourism Type.

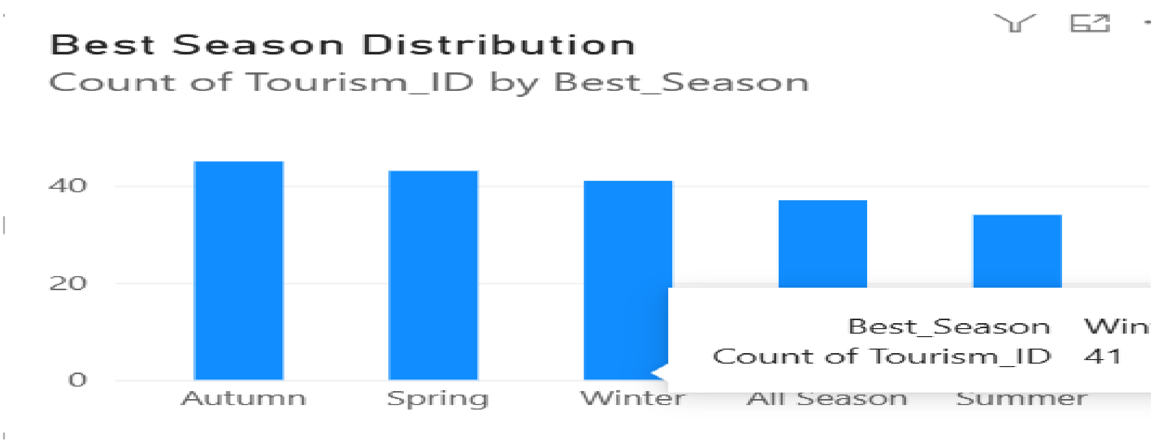
   - Add a Bar Chart for Annual Visitors by Continent.



- Add a Chart for Average Budget and Average Duration by Tourism Type.



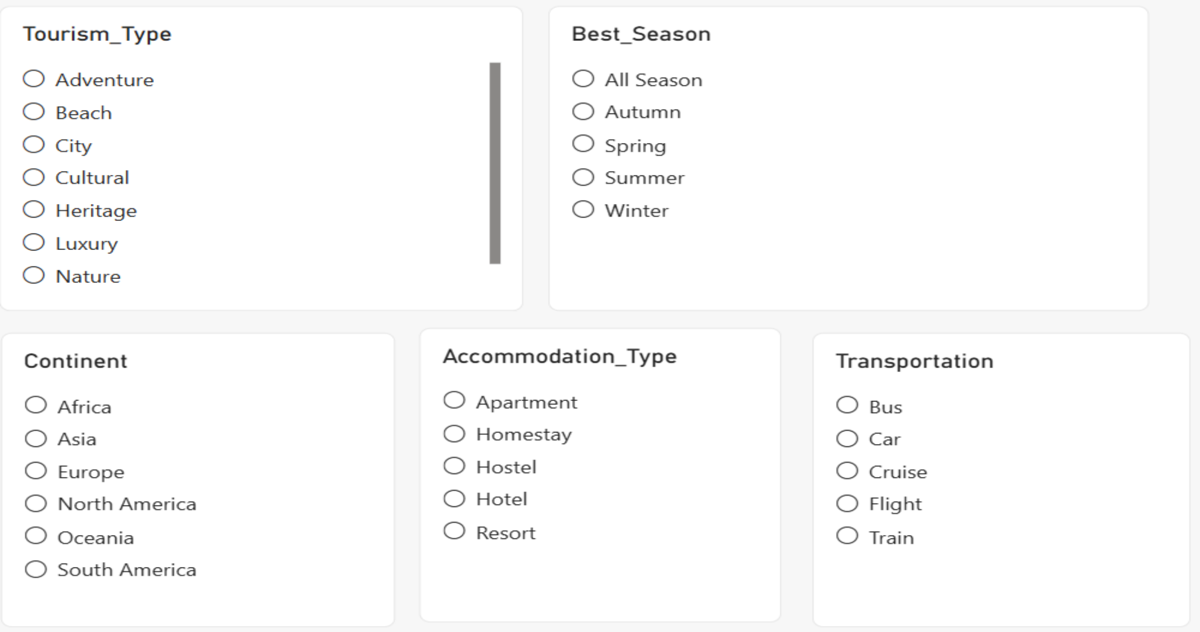
- Add a Chart for Best Season Distribution.



• Add Slicers for:

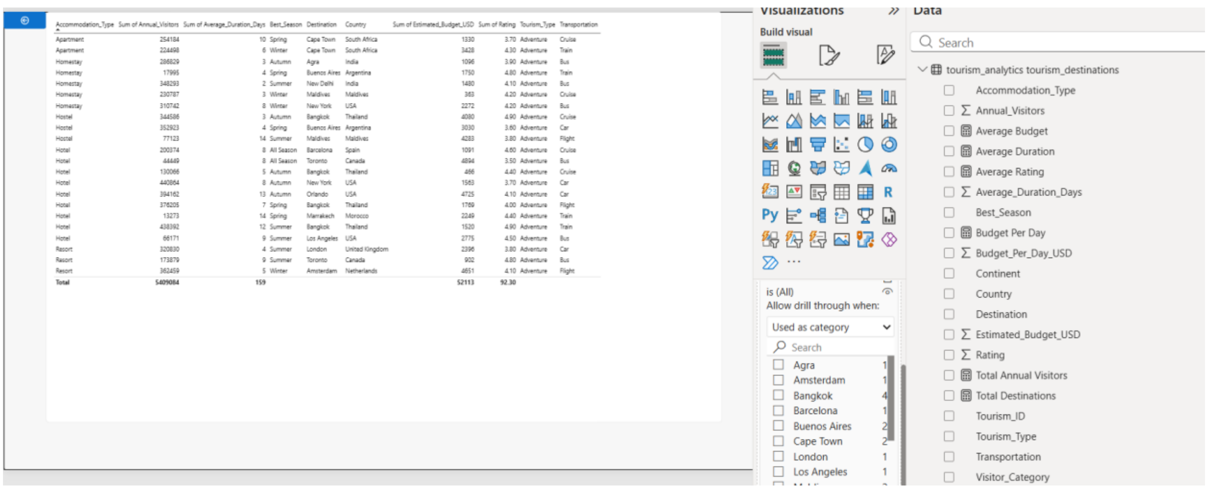
- Continent

- Tourism Type
- Best Season
- Accommodation Type
- Transportation



2. **Destination Details Drill-Through Page:**
   - Create a second Power BI page called:

   - **Destination Details**
   - Add **Destination** to the **Drill-through** field.
   - Then add a table containing:

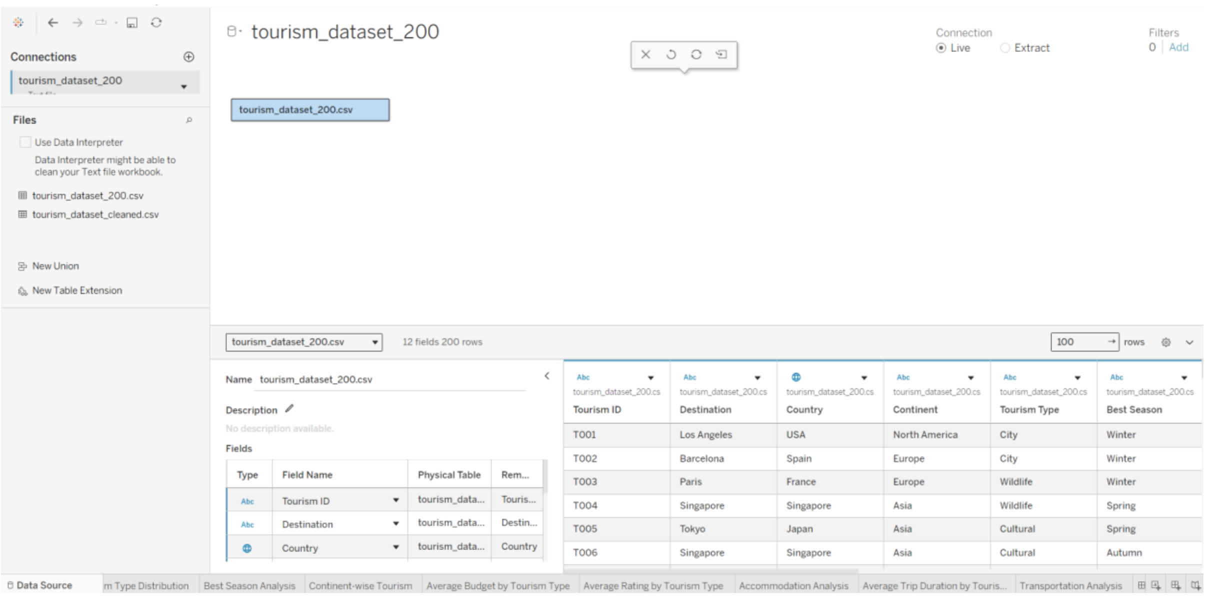


Step 6: Tableau Data Connection & Preparation

1. Open Tableau Desktop / Tableau Public.

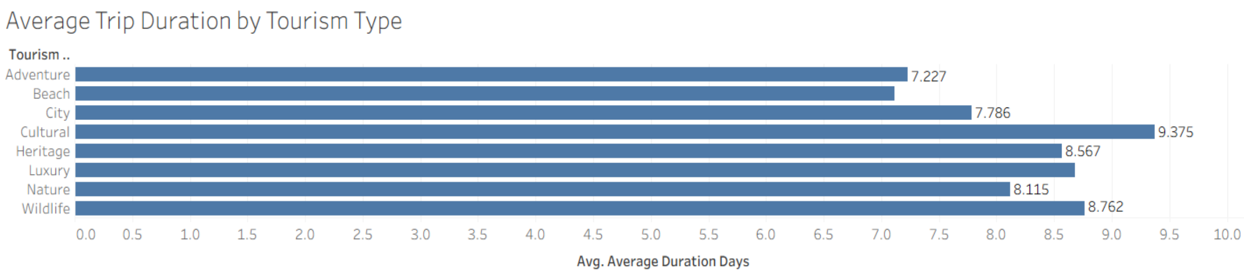
2. Under Connect, select To a File → Text File and choose `tourism_dataset_200.csv`.

3. Verify the tourism dataset fields and data types, then load the dataset into the Tableau worksheet.



4. Go to any sheet and build calculated fields:

- **Budget Per Day**
  [Estimated Budget USD] / [Average Duration Days]
- **Visitors in Millions**
  [Annual Visitors] / 1000000
- **Destination Count**
  COUNTD([Tourism ID])

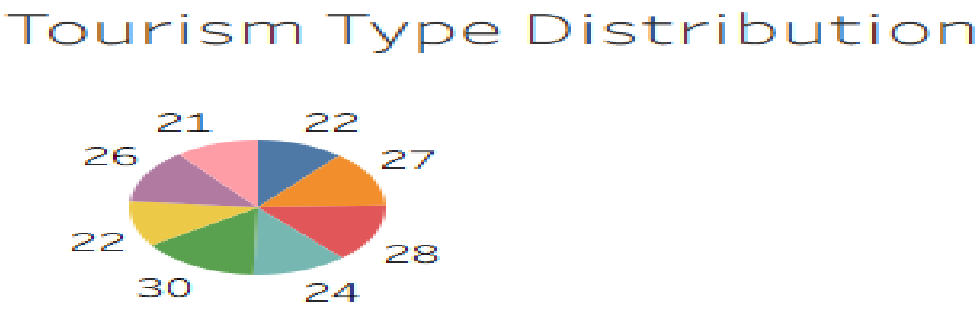


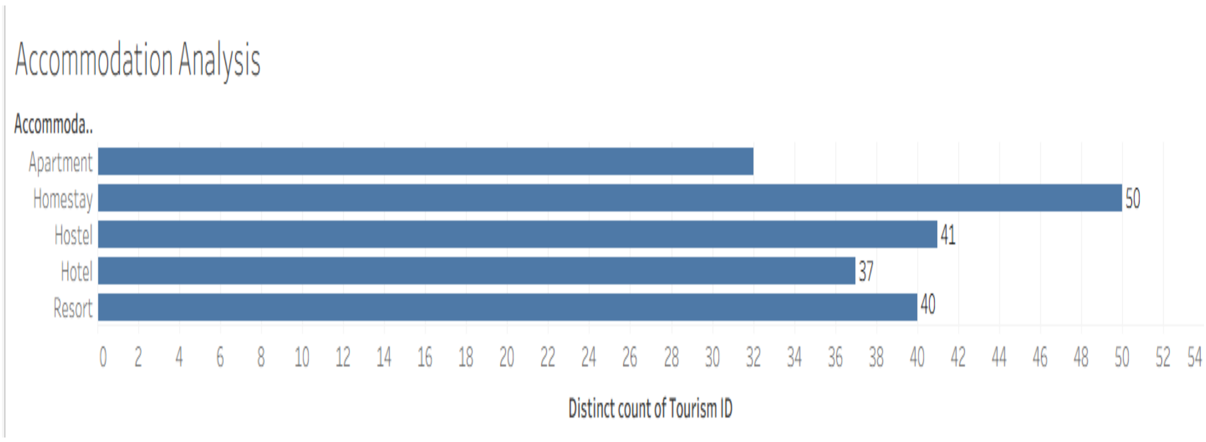
Step 7: Tableau Companion Dashboard & Interactive Story

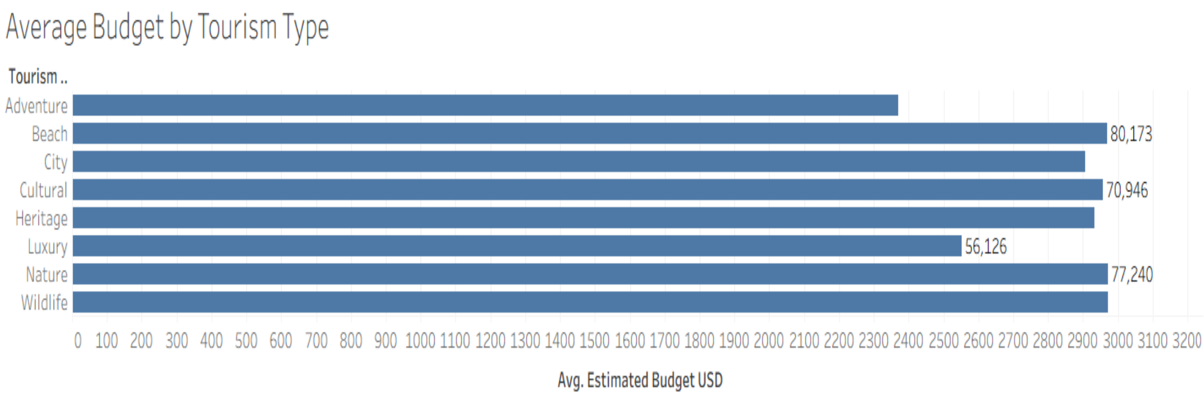
1. Worksheets: Build the following visualizations:

- **Tourism Type Distribution** – Pie Chart
- **Best Season Analysis** – Bar Chart
- **Continent-wise Tourism** – Bar Chart

- **Average Budget by Tourism Type** – Bar Chart
- **Average Rating by Tourism Type** – Bar Chart
- **Average Trip Duration by Tourism Type** – Bar Chart
- **Accommodation Analysis** – Bar Chart
- **Transportation Analysis** – Bar Chart



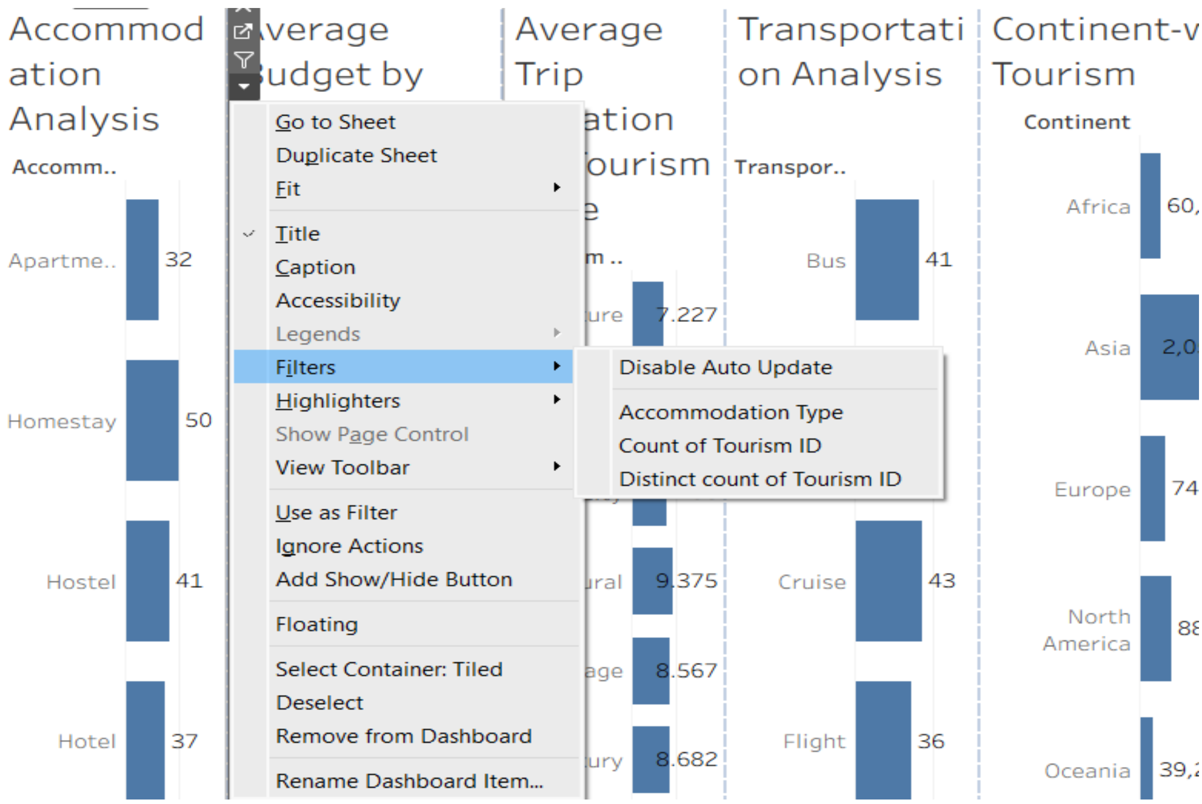




2. Global Filters & Actions:

- Drag **Continent**, **Tourism Type**, **Best Season**, **Accommodation Type**, and **Transportation** to the **Filters** shelf.
- Set the filters to apply to **all worksheets using this data source**.
- Create a new **Dashboard** tab and drag the important worksheets onto the dashboard.
- Use the **Tourism Type Distribution** or **Continent-wise Tourism** chart as a filter so that selecting a category updates the other visualizations.

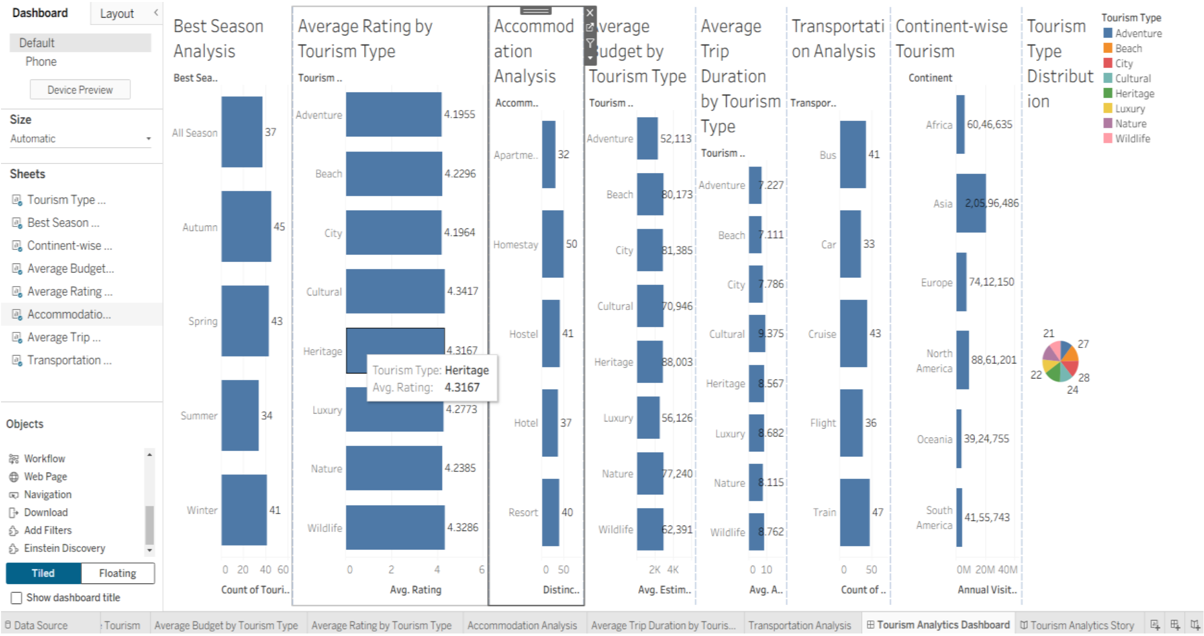
- Add tooltips showing **Destination, Country, Tourism Type, Rating, Estimated Budget, Average Duration, and Annual Visitors**.



3. Dashboard Layout:

Arrange the charts to create an interactive tourism analytics dashboard with:

- Tourism Type Distribution
- Visitor Distribution by Continent
- Best Season Analysis
- Average Budget
- Average Rating
- Average Duration
- Accommodation and Transportation analysis



Observed Insights:

- Dataset Overview: The tourism dataset contains 200 destination records across 21 countries, 6 continents, and 8 tourism types. The overall average rating is 4.26, average estimated budget is $2,841.89, and average duration is 8.19 days.
- Tourism Type Distribution: Heritage has the largest number of destination records (30), followed by City (28) and Beach (27).
- Visitor Demand: Asia is the leading continent by total annual visitors, with 20,596,486 annual visitors represented in the dataset.
- Seasonality: Autumn has the highest number of destination records (45) and the highest total annual visitors among the season categories in this dataset.
- Tourism Experience: Cultural has the highest average rating among the tourism types at 4.34. Cultural has the longest average trip duration at 9.38 days.
- Travel Services: Homestay is the most common accommodation type (50 records), while Train is the most common transportation mode (47 records).